CATERPILLAR®

DEALER BEST PRACTICE SERIES

Optimizing Component Removal & Installation Quality through Customer Certification

Application | Maintenance | Site Management | **Component Rebuild** | Safety | MARC Management

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1.0 Introduction

Proper Major Component Removal and Installation (R&I) practices are vitally important for component reliability and life. Mistakes made during component removal and installations are a leading cause of early hour failures or shortened life. To ensure that component removal and installation is performed following a well-defined process, many Caterpillar dealers have provided training, and in some cases, even “Certified” customers as qualified to perform component R & I.

2.0 Best Practice Description

Caterpillar Dealers have invested considerable financial and manpower resources into creating major component rebuild capabilities that deliver cost efficient, high quality for their customers. A poorly performed removal and / or installation process can negate the efforts put into component rebuild.

Some dealers have taken steps to control the R&I process in order to protect their investment in the rebuild and also to ensure that the component delivers expected reliability and life.

Typical actions have included:

- Dealer must do the component installation.
- Dealer must supervise the installation
- Dealer trains and certifies the customer to perform component removals and installations.

3.0 Implementation Steps

Start with a process audit by the dealer to the customer. Then provide subsequent training on proper major component removal and installation. The most important success factor is a common, shared objective by customer and dealer. A shared objective will ensure component reliability and life through proper R & I practices.

Implementation Steps:

1. **Customer Shop Audit / Inspection**
    a. Tooling
    b. Contamination control practices
    c. Cleaning equipment
    d. R & I area
    e. Component storage / staging

2. **Training:**
   a. Failure Analysis – Determine cause for removal
   b. Document removal – hours, history, oil analysis, particle count
   c. System inspection and clean up process– debris removal
   d. Rebuild related system: radiator, coolers, pumps, hoses, and air intake, etc.
   e. Installation procedures / checklists
   f. Test and brake-in procedures
   g. Fluid cleanliness - ISO particle count specifications
   h. Standardized installation parts kits (see example to the right)
   i. Record keeping –installation checklists, parts Bill of Materials, particle count, test results.
   j. Dealer feedback – completed installation checklists, initial SOS

Install Kit – 793C Final Drive

SERIAL	MODEL	PART N.	DESCRIPTION	QTY
4AR	793C	1R0719	FILTER A	1
4AR	793C	2D6506	SEAL O RING	1
4AR	793C	2M9780	SEAL O RING	3
4AR	793C	2S8439	SEAL O RING	1
4AR	793C	3J1901	SEAL	4
4AR	793C	5P1265	CM-HOSE STK	77
4AR	793C	5P1935	SEAL	2
4AR	793C	6V8398	SEAL O RING	3
4AR	793C	7X4801	SEAL	1
4AR	793C	8W3192	HOSE A	1
4AR	793C	8X8172	HOSE A	1
4AR	793C	8X8173	HOSE A	1
4AR	793C	9C4937	BREATHER A	1
4AR	793C	9R2587	HOSE AS	1
4AR	793C	9X8499	WASHER	47
4AR	793C	0068352	RING	2
4AR	793C	1293175	BOLT-HEXA01	41
4AR	793C	1313425	HOSE	1
4AR	793C	1440367	CLAMP	4
4AR	793C	1779754	BOLT	3
4AR	793C	1779755	BOLT	3

OHT COMPONENTS AND SYSTEMS

ENGINE	TORQUE CONVERTER	TRANSMISSION
Cooling System	Brake Oil Coolers	Charge Pump
Radiator	Filters	Lube Pump
Coolant Hoses / Clamps	Drive Shaft	Scavenge Pump
Oil Cooler Hoses	U-Joints / Bolts	Brake Cooling Pump
Air Intake System	Hoses	Oil Cooler
Filters		T/C Lock-up Clutch Valves
Ducts		Hoses
Tubes / Clamps		Filter
Air Compressor		Harness
Hoses		
Harness		
Engine Mounts		

During replacement of any major component, attention should be given to the related systems & sub-components, which may impact performance and life of the newly replaced component.

4.0 Benefits

- **Improved reliability**
  - Improper installation is a leading cause of early hour component failures and shortened life.
  - Proper installation positively impacts:
    - Machine availability, which impacts production rates at the site.
    - Maintenance costs by avoiding unnecessary costly repairs.

- **Improved durability**.
  - Proper installation helps to maximize component life and reduce cost per ton.

5.0 Resources Required

- Dealer must have a qualified inspector and training instructor. In addition, training materials may need to be developed.

- Customer may require improved tooling or shop facility improvements.

6.0 Supporting Attachments / References

References:

See also, *Improving Component Durability – Component Removal and Installation -* SEBF1017

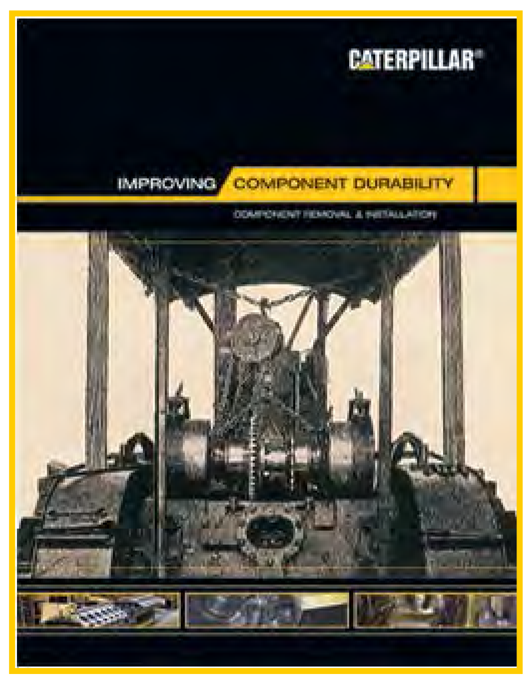
This booklet explains how problems in the component removal and installation process often cause components to fail.

The booklet contains 24 pages of high -quality, full color graphics and text, which provide an easy -to -understand explanation of:

- *Common failures caused by poor R&I practices*
- *Importance of the R&I process*
- *Best practices*
- *Risk management in cleaning contaminated systems*

The booklet also explains the how the component replacement process has evolved from the days of the early track-type tractors to today's modern machines with electronically controlled engines and transmissions. A common sense approach to risk management is also discussed regarding how much time to invest in system disassembly and cleaning after a catastrophic component failure.

The booklet is intended for all levels of dealer and customer personnel involved with the operation and maintenance of earthmoving equipment. It is particularly useful to those who manage equipment operation and maintenance.



Attachments:
R&I Checklists in SIS (pdf).zip. The attached .zip file contains a collection of R&I checklists for various equipment models. Please click on the on the *"Attachments"* tab in this electronic Adobe file.

Removal & Installation Checklist - Shovels/Excavators			
Caterpillar Form No.	01-040500-00 (PC)	Models 5110, 5130, 5230	- Engine
Caterpillar Form No.	01-040501-00 (PC)	Models 5110, 5130, 5230	- Final Drive
Caterpillar Form No.	01-040502-00 (PC)	Models 5110, 5130, 5230	- Swing Drive
Caterpillar Form No.	01-040503-00 (PC)	Models 5110, 5130, 5230	- Drive Pump
Caterpillar Form No.	01-040504-00 (PC)	Models 5110, 5130, 5230	- Swing Pump
Caterpillar Form No.	01-040505-00 (PC)	Model 5110	- Implement Pump
Caterpillar Form No.	01-040506-00 (PC)	Model 5130	- Implement Pump
Caterpillar Form No.	01-040507-00 (PC)	Model 5230	- Implement Pump
Removal & Installation Checklist - Motor Graders			
Caterpillar Form No.	01-040508-00 (PC)	Model 14 & 16	- Engine
Caterpillar Form No.	01-040509-00 (PC)	Model 14 & 16	- Transmission & Differential
Caterpillar Form No.	01-040510-00 (PC)	Model 24	- Differential
Caterpillar Form No.	01-040511-00 (PC)	Model 24	- Engine & Torque Converter
Caterpillar Form No.	01-040512-00 (PC)	Model 24	- Torque Converter
Caterpillar Form No.	01-040513-00 (PC)	Model 24	- Transmission
Removal & Installation Checklist - Off Highway Trucks			
Caterpillar Form No.	01-040514-00 (PC)	Models 777, 785, 789, 793	- Differential
Caterpillar Form No.	01-040515-00 (PC)	Models 777, 785, 789, 793	- Engine & Torque Converter
Caterpillar Form No.	01-040516-00 (PC)	Models 777, 785, 789, 793	- Final Drive
Caterpillar Form No.	01-040517-00 (PC)	Models 777, 785, 789, 793	- Torque Converter
Caterpillar Form No.	01-040518-00 (PC)	Models 777, 785, 789, 793	- Transmission
Caterpillar Form No.	01-040519-00 (PC)	Model 797	- Differential
Caterpillar Form No.	01-040520-00 (PC)	Model 797	- Engine & Torque Converter
Caterpillar Form No.	01-040521-01 (PC)	Model 797	- Final Drive
Caterpillar Form No.	01-040522-00 (PC)	Model 797	- Torque Converter
Caterpillar Form No.	01-040523-00 (PC)	Model 797	- Transmission
Removal & Installation Checklist - Track Type Tractors			
Caterpillar Form No.	01-040524-00 (PC)	Models D9, D10, D11	- Engine & Torque Divider
Caterpillar Form No.	01-040525-00 (PC)	Models D9, D10, D11	- Final Drive and Steering Clutch & Brake
Caterpillar Form No.	01-040526-00 (PC)	Models D9, D10, D11	- Torque Divider
Caterpillar Form No.	01-040527-00 (PC)	Models D9, D10, D11	- Transmission & Bevel Gear
Removal & Installation Checklist - Wheel Loaders & Dozers			
Caterpillar Form No.	01-040528-00 (PC)	Models 992, 994, 834, 844, 854	- Engine & Torque Converter
Caterpillar Form No.	01-040529-01 (PC)	Models 992, 994, 834, 844, 854	- Final Drive & Service Brake
Caterpillar Form No.	01-040530-00 (PC)	Models 992, 994, 834, 844, 854	- Differential
Caterpillar Form No.	01-040531-00 (PC)	Models 992, 994, 834, 844, 854	- Torque Converter
Caterpillar Form No.	01-040532-00 (PC)	Models 992, 994, 834, 844, 854	- Transmission

7.0 Related Best Practices
None applicable.

8.0 Acknowledgements

This Best Practice was written by:

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